

SESSION 1 LAB EXPERIMENTS

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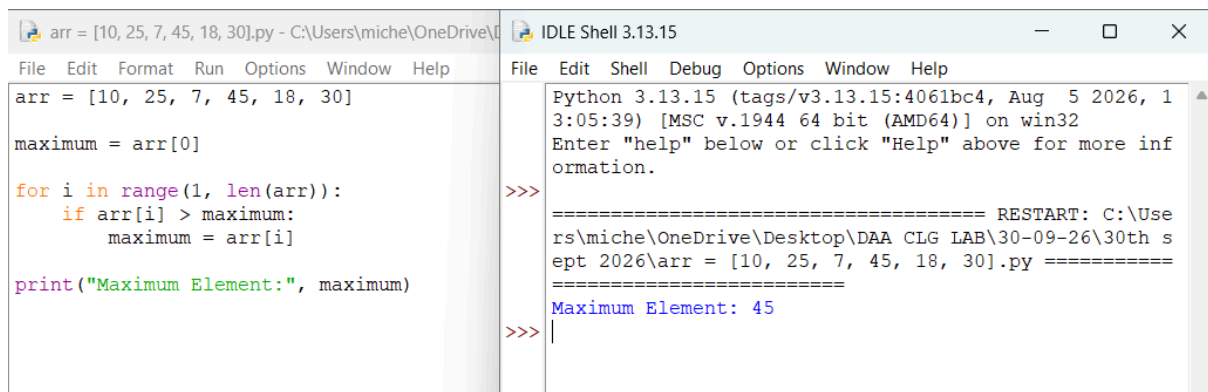
1.

```
arr = [10, 25, 7, 45, 18, 30]
```

```
maximum = arr[0]
```

```
for i in range(1, len(arr)):
    if arr[i] > maximum:
        maximum = arr[i]
```

```
print("Maximum Element:", maximum)
```



The screenshot shows a Python IDE window titled 'arr = [10, 25, 7, 45, 18, 30].py - C:\Users\miche\OneDrive\...' and 'IDLE Shell 3.13.15'. The left pane contains the Python code for finding the maximum element in an array. The right pane shows the output of the script, which is 'Maximum Element: 45'.

```
arr = [10, 25, 7, 45, 18, 30]
maximum = arr[0]
for i in range(1, len(arr)):
    if arr[i] > maximum:
        maximum = arr[i]
print("Maximum Element:", maximum)
```

```
Python 3.13.15 (tags/v3.13.15:4061bc4, Aug 5 2026, 13:05:39) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
===== RESTART: C:\Users\miche\OneDrive\Desktop\DAA CLG LAB\30-09-26\30th s
ept 2026\arr = [10, 25, 7, 45, 18, 30].py =====
Maximum Element: 45
```

2.

```
arr1 = [1, 2, 3, 4, 5]
```

```
arr2 = [3, 4, 5, 6, 7]
```

```
common = []
```

```
for x in arr1:
    if x in arr2:
        common.append(x)
```

```
print("Common Elements:", common)
```

```
Common in two arrays.py - C... IDLE Shell 3.13.15
File Edit Format Run Options Window Help File Edit Shell Debug Options Window Help
arr1 = [1, 2, 3, 4, 5]
arr2 = [3, 4, 5, 6, 7]

common = []

for x in arr1:
    if x in arr2:
        common.append(x)

print("Common Elements:", common)

Python 3.13.15 (tags/v3.13.15:4061bc4, Aug 5 2026, 13:05:39) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

= RESTART: C:\Users\miche\OneDrive\Desktop\DAA CLG LAB\30-09-26\30th sept 2026\Common in two arrays.py
Common Elements: [3, 4, 5]
```

3.

```
arr = [3, 1, 2, 3, 1, 3]
```

```
k = 2
```

```
count = 0
```

```
for i in range(len(arr)):
```

```
    for j in range(i + 1, len(arr)):
```

```
        if arr[i] == arr[j] and (i * j) % k == 0:
```

```
            count += 1
```

```
print("Number of valid pairs:", count)
```

```
Count Equal and Divisible Pairs in an Array.py - C:\Users\miche\OneDrive Desktop IDLE Shell 3.13.15
File Edit Format Run Options Window Help File Edit Shell Debug Options Window Help
arr = [3, 1, 2, 3, 1, 3]
k = 2

count = 0

for i in range(len(arr)):
    for j in range(i + 1, len(arr)):
        if arr[i] == arr[j] and (i * j) % k == 0:
            count += 1

print("Number of valid pairs:", count)

Python 3.13.15 (tags/v3.13.15:4061bc4, Aug 5 2026, 13:05:39) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

= RESTART: C:\Users\miche\OneDrive\Desktop\DAA CLG LAB\30-09-26\30th sept 2026\Count Equal and Divisible Pairs in an Array.py
Number of valid pairs: 3
```

4.

```
arr = ["apple", "hello", "madam", "level", "world"]
```

```
for word in arr:
```

```
    if word == word[::-1]:
```

```
        print("First Palindromic String:", word)
```

```
        break
```

```
Palindrome.py - C:\Users\miche\OneDrive\Desktop\DAA CLG LAB\30-09-2026\
File Edit Format Run Options Window Help
arr = ["apple", "hello", "madam", "level", "world"]

for word in arr:
    if word == word[::-1]:
        print("First Palindromic String:", word)
        break

IDLE Shell 3.13.15
File Edit Shell Debug Options Window Help
Python 3.13.15 (tags/v3.13.15:4061bc4, Aug 5 2026, 13:05:39) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> = RESTART: C:\Users\miche\OneDrive\Desktop\DAA CLG LAB\30-09-26\30th sept 2026\Palindrome.py
First Palindromic String: madam
>>>
```

5.

```
arr = [1, 2, 1]
```

```
total = 0
```

```
for i in range(len(arr)):
    distinct = set()
```

```
    for j in range(i, len(arr)):
        distinct.add(arr[j])
        total += sum(x * x for x in distinct)
```

```
print("Sum of squares of distinct elements:", total)
```

```
Subarrays Distinct Element Sum of Squares.py - C:\Users\miche\OneDrive\Desktop\
File Edit Format Run Options Window Help
arr = [1, 2, 1]

total = 0

for i in range(len(arr)):
    distinct = set()

    for j in range(i, len(arr)):
        distinct.add(arr[j])
        total += sum(x * x for x in distinct)

print("Sum of squares of distinct elements:", total)

IDLE Shell 3.13.15
File Edit Shell Debug Options Window Help
Python 3.13.15 (tags/v3.13.15:4061bc4, Aug 5 2026, 13:05:39) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> = RESTART: C:\Users\miche\OneDrive\Desktop\DAA CLG LAB\30-09-26\30th sept 2026\Subarrays Distinct Element Sum of Squares.py
Sum of squares of distinct elements: 21
>>>
```